Binomial Expression Expansion

Example: $(x+y)^3 = \binom{3}{0}x^3 + \binom{3}{1}x^2y + \binom{3}{2}xy^2 + \binom{3}{3}y^3$

$$(x+y)(x+y)(x+y)$$

$$= xxx + xyx + yxx + yyx + xxy + xyy + yxy + yyy.$$

Coefficient of xy^2

= number of length-3 lists having y in 2 places.

You can think of x as 0 and y as 1 in a bitstring

The equivalent question is how many bit strings have exactly 2 1's: C(3,2).